# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 1

**Full Syllabus (Chapters 1–14)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** The point $(5, -2)$ is reflected in the $y$-axis. The image is: 1

**(a)** $(-5, 2)$ **(b)** $(-5, -2)$ **(c)** $(-2, -5)$ **(d)** $(5, -2)$

**2.** Which of these is a linear polynomial? 1

**(a)** $x^2 + 4$ **(b)** $x^3 - x$ **(c)** $7$ **(d)** $3x + 4$

**3.** The number of decimal places in the expansion of $\frac{3}{2^3 \times 5^2}$ is: 1

**(a)** 5 **(b)** 2 **(c)** 3 **(d)** 4

**4.** $(2x + 3y)^2 =$ 1

**(a)** $4x^2 + 6xy + 9y^2$ **(b)** $4x^2 + 12xy + 9y^2$ **(c)** $4x^2 - 12xy + 9y^2$ **(d)** $4x^2 + 9y^2$

**5.** A chord is at distance $7$ cm from the centre of a circle of radius $25$ cm. Its length (in cm) is: 1

**(a)** 47 **(b)** 48 **(c)** 49 **(d)** 50

**6.** The area of a triangle with sides $13$ cm, $14$ cm and $15$ cm is (cm$^2$): 1

**(a)** 85 **(b)** 83 **(c)** 84 **(d)** 86

**7.** A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 2 is: 1

**(a)** $\frac{6}{25}$ **(b)** $\frac{11}{50}$ **(c)** $\frac{11}{51}$ **(d)** $\frac{39}{50}$

**8.** If $t_n = 2n^2 + 3$, then $t_5$ is: 1

**(a)** $54$ **(b)** $53$ **(c)** $52$ **(d)** $55$

**9.** The converse of the proposition “If a number is divisible by 6, then it is divisible by 3” is: 1

**(a)** If it is not true that it is divisible by 3, then a number is divisible by 6.
**(b)** If a number is divisible by 6, then it is divisible by 3.
**(c)** If a number is divisible by 3, then it is divisible by 6.
**(d)** If it is not true that a number is divisible by 6, then it is not true that it is divisible by 3.

**10.** The average marks of 27 students is $57$ and of another 39 students is $35$. The average of all 66 students is: 1

**(a)** $43$ **(b)** $45$ **(c)** $46$ **(d)** $44$

**11.** Adding $1747$ and $4333$ by the column algorithm, the number of columns in which a carry is generated is: 1

**(a)** $2$ **(b)** $3$ **(c)** $4$ **(d)** $1$

**12.** The angles of a quadrilateral are in the ratio $1:2:6:6$. The largest angle is: 1

**(a)** $144^\circ$ **(b)** $164^\circ$ **(c)** $154^\circ$ **(d)** $134^\circ$

**13.** Is the point $(-3, 2)$ a solution of $x + 2y = 2$? 1

**(a)** Only for $x = 0$ **(b)** Yes **(c)** Cannot be said **(d)** No

**14.** Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius $3.5$ cm and height $11$ cm is: 1

**(a)** $242$ cm$^2$ **(b)** $267$ cm$^2$ **(c)** $217$ cm$^2$ **(d)** $292$ cm$^2$

**15.** The distance between $(1, 2)$ and $(4, 6)$ is: 1

**(a)** 6 **(b)** 5 **(c)** 7 **(d)** 4

**16.** If $x = 5$ is a zero of the polynomial $2x + k$, then $k =$ 1

**(a)** -9 **(b)** -11 **(c)** -8 **(d)** -10

**17.** $x = 0.\overline{27}$ equals: 1

**(a)** $\frac{2}{11}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{1}{4}$

**18.** $(3x - 2y)^2 =$ 1

**(a)** $9x^2 - 6xy + 4y^2$ **(b)** $9x^2 + 4y^2$ **(c)** $9x^2 - 12xy + 4y^2$ **(d)** $9x^2 + 12xy + 4y^2$

**19.** The radius of a circle is $25$ cm and a chord is $40$ cm long. The distance of the chord from the centre (in cm) is: 1

**(a)** 16 **(b)** 15 **(c)** 14 **(d)** 17

**20.** The semi-perimeter of a triangle with sides $13$ cm, $14$ cm and $15$ cm is (cm): 1

**(a)** 22 **(b)** 21 **(c)** 23 **(d)** 20

## SECTION — B (Very Short Answer Type) — 6 × 2 = 12

*Answer in about 20–30 words each.*

**21.** Find the distance between $P(2, -1)$ and $Q(5, 3)$. 2

**22.** Find the zero of $3x - 18$ and show that it lies on the $x$-axis of the graph $y = 3x - 18$. 2

**23.** Express $0.\overline{3}$ as a fraction in lowest terms. 2

**24.** Expand $(3x + 5y)^2$. 2

**25.** In a circle of radius $26$ cm, a chord is $10$ cm from the centre. Find its length. 2

**26.** Find the area of a triangle with sides $13$ cm, $14$ cm, $15$ cm. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 40–60 words each.*

**27.** Show that the points $A(0, 0)$, $B(4, 3)$ and $C(8, 0)$ are the vertices of an isosceles triangle. 3

**28.** Draw up a table for $y = 2x + 3$ for $x = -1, 0, 1, 2$ and state the slope and the $y$-intercept. 3

**29.** Express $0.\overline{36}$ in the form $\frac{p}{q}$. 3

**30.** Expand $(2x + 3y)^2$ and verify for $x = 1$, $y = 1$. 3

**31.** In triangle $OAB$ with $O$ the centre of a circle, $\angle AOB = 70^\circ$. Find $\angle OAB$ and the angle subtended by $AB$ at a point of the major arc. 3

**OR**

Find the area of the triangle with sides $13$ cm, $14$ cm and $15$ cm, and the length of the altitude on the side of $15$ cm. 3

**32.** A bag contains 4 red, 3 black and 5 white balls. A ball is drawn at random. Find the probability that it is 3
(i) red,
(ii) not red,
(iii) black or white.

**OR**

Find which term of the AP $7, 13, 19, 25, \ldots$ equals $151$. 3

## SECTION — D (Long Answer Type) 6 × 5 = 30

*Answer in about 100–150 words each.*

**33.** Show that the points $A(0, 0)$, $B(4, 1)$, $C(6, 4)$ and $D(2, 3)$ taken in order form a parallelogram. 5

**34.** The cost of $x$ notebooks is a linear polynomial: a shop charges ₹60 for the bag plus ₹12 per notebook. Write the cost $C(x)$, find the cost of 5 notebooks and how many notebooks can be bought for ₹168. 5

**35.** Find the decimal expansion of $\frac{1}{7}$ by long division and state the length of the repeating block. 5

**36.** Expand $(2x - 3y)^2$ and find its value when $x = 3$ and $y = 1$. 5

**OR**

Two parallel chords of lengths $10$ cm and $24$ cm lie on the same side of the centre of a circle of radius $13$ cm. Find the distance between them. 5

**37.** A circular track has inner radius $14$ m and outer radius $21$ m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹5 per m$^2$. 5

**OR**

A die was thrown 100 times; the number of times $1, \ldots, 6$ appeared was 15, 22, 17, 16, 14, 16. Find the experimental probability of 5
(i) a prime number,
(ii) a number greater than 4,
(iii) a number less than 3, and check that all six relative frequencies add to $1$.

**38.** A taxi charges a fixed ₹200 and ₹40 per km. 5
(i) Show that the fares for 1, 2, 3 km form an AP.
(ii) Find the fare for 12 km.

**OR**

Prove that the sum of any 3 consecutive integers is divisible by 3. Verify for the numbers starting at $8$. 5

**END OF QUESTION PAPER** **Total: 80 Marks**

NEET MIND